

keypoints_pipeline - Conversation Log

get context of the directory : keypoints_pipeline and generate a requirements.txt and readme.md for github push

Agent:Explore keypoints_pipeline directory

IN

Explore the directory c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline thoroughly. I need to understand:

1. All Python files and their imports (to derive requirements)
2. The overall purpose and architecture of the pipeline
3. Key scripts, their roles, and how they connect
4. Any config files, existing docs, or README/requirements that exist
5. Model architectures used (ResNet, EfficientNet, custom CNN etc.)
6. Dataset structure and data flow
7. Training, evaluation, and inference scripts

Please read ALL Python files and any markdown/text files. List all third-party imports found across all files so I can generate an accurate requirements.txt. Be very thorough.

Bash

IN

```
find "c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline" -type f -name "*.py" -o -name "*.md" -o -name "*.txt" -o -name "*.json" -o -name "*.yaml" -o -name "*.yml" 2>/dev/null | head -100
```

OUT

Exit code 1

Bash

IN

```
cd /c:/Users/sudanb/Desktop/CV_datasets/keypoints_pipeline && find . -type f \( -name "*.py" -o -name "*.md" -o -name "*.txt" -o -name "*.json" -o -name "*.yaml" -o -name "*.yml" \) 2>/dev/null | sort
```

OUT

Exit code 1

Bash

IN

```
ls -la "/c:/Users/sudanb/Desktop/CV_datasets/keypoints_pipeline" 2>&1
```

OUT

Exit code 2

Bash

IN

```
dir "c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline" 2>&1 | head -30
```

OUT

Exit code 1

Globpattern: "**/*.py"

Found 12 files

Globpattern: "**/*.yaml"

No files found

Read c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\train_all.py

Read c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\train_custom.py

Read c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\training_data_fmt_17\mmdet_dataset_config.py

Read c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\keypoints_2d.json

Read c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\training_data_fmt_17\maize.yaml

Read c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\checkpoints\efficientnet_b3\history.json

Bash

IN

```
find /c/Users/sudanb/Desktop/CV_datasets/keypoints_pipeline -maxdepth 2 -type f \( -name "*.md" -o -name "*.rst" -o -name "*.txt" \) 2>/dev/null | grep -v label
```

OUT

(Bash completed with no output)

Grep "^import |^from " (in c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline, type: py)

56 lines of output

Bash

IN

```
cd /c/Users/sudanb/Desktop/CV_datasets/keypoints_pipeline && find . -type d -maxdepth 2 | sort
```

OUT

Exit code 1

Globpattern: "***"

Found 101 files

I now have a complete picture of the pipeline. Let me read a couple of key files to confirm exact imports before writing the files.

Globpattern: "keypoints_pipeline/*.py"

Found 11 files

Read c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\measure_plants.py

Read c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\project_keypoints.py

Read c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\visualize_suspicious.py

Read c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\reg_dataset.py

Read c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\reg_model.py

Now let me check the existing methodology.md and any eval results for the README.

Read c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\methodology.md

Read c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\checkpoints\efficientnet_b3\eval_results.json

Now I have everything I need. Writing both files now.

Write c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\requirements.txt

8 lines

torch>=2.0.0

torchvision>=0.15.0

numpy>=1.24.0

scipy>=1.10.0

Pillow>=9.0.0

plyfile>=0.8.1

opencv-python>=4.7.0

Write c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\README.md

211 lines

Maize Plant Phenotyping - Keypoints Pipeline

End-to-end pipeline for predicting 3D plant measurements (stem length, internode lengths, leaf lengths) from single RGB images of field-grown maize (*Zea mays*). The pipeline covers 3D point cloud processing, 2D keypoint projection, and deep regression model training/evaluation.

Overview

...

Raw PLY point clouds

```

  |
  v
measure_plants.py  -- Extract stem, internode, and leaf lengths via MST
  |
  v
project_keypoints.py  -- Project 3D attachment keypoints onto 2D RGB images
  |
  v
reg_dataset.py      -- Build PyTorch Dataset (640x640 images -> 32 targets)
  |
  v
train_*.py          -- Train EfficientNet-B3 / ResNet-34 / Custom CNN
  |
  v
eval_reg.py         -- Evaluate single-view and multi-view (75 views/plant)

```

...

Dataset

- **Source**: Field-grown maize point clouds (segmented PLY files, 100k points/plant)
- **Plants**: ~520 plants, up to 16 leaves per plant
- **Camera**: Intel RealSense D435 (640x640, hfov=90deg, vfov=65deg)
- **Views per plant**: 75 (5 azimuths x 5 elevations x 3 distances)
- **Total images**: ~39 000
- **Split**: 70% train / 15% val / 15% test

Regression targets (32 per sample)

Index	Target
0	Stem length (m)
1-15	Internode lengths 1-15 (m)
16-31	Leaf lengths 1-16 (m)

Missing targets (e.g. plant has only 10 leaves) are masked out of the loss.

Models

All models share the same regression head:

...

```

Linear(in_dim, 512) -> BN -> ReLU -> Dropout
Linear(512, 128)   -> BN -> ReLU -> Dropout

```

```
Linear(128, 32)  -> ReLU      # non-negative outputs
```

```
...
```

```
| Model | Encoder | Feature dim | Trainable from |
|-----|-----|-----|-----|
| `efficientnet` | EfficientNet-B3 | 1536 | ImageNet pretrained |
| `resnet34` | ResNet-34 (to layer 3) | 512 | ImageNet pretrained |
| `custom` | Custom CNN (residual blocks) | 512 | Scratch |
```

****Training details****

- Optimizer: AdamW (separate LR for encoder and head)
- Scheduler: CosineAnnealingLR (eta_min = 1e-6)
- Loss: Masked MSE
- Gradient clipping: max_norm = 1.0
- Augmentation: horizontal flip, rotation +/-10deg, color jitter, Gaussian blur, random erasing

```
---
```

Installation

```
``bash
git clone <repo-url>
cd keypoints_pipeline
pip install -r requirements.txt
...
```

> ****Note****: Install PyTorch matching your CUDA version from [pytorch.org](https://pytorch.org/get-started/locally/) before running pip install.

```
---
```

Usage

1. Measure plants from point clouds

```
``bash
python measure_plants.py
...
```

Outputs `labels.json` (stem/internode/leaf lengths) and `keypoints.json` (3D attachment coordinates). Edit `SEG\_DIR` at the top of the file to point to your segmented PLY directory.

2. Project keypoints to 2D

```
``bash
python project_keypoints.py
...
```

Outputs `keypoints\_2d.json` with per-image 2D keypoint coordinates and visibility flags.

3. Train models

Train all three models sequentially:

```
```bash
python train_all.py
```
```

Or train individually:

```
```bash
python train_efficientnet.py
python train_resnet34.py
python train_custom.py
```
```

Checkpoints are saved to `checkpoints/{efficientnet\_b3,resnet34,custom\_cnn}/best.pt`.

4. Evaluate

```
```bash
python eval_reg.py --model efficientnet --ckpt checkpoints/efficientnet_b3/best.pt
python eval_reg.py --model resnet34 --ckpt checkpoints/resnet34/best.pt
python eval_reg.py --model custom --ckpt checkpoints/custom_cnn/best.pt
```
```

Reports single-view and multi-view (75-view average) MAE, RMSE, and R2 per slot and overall. Saves per-plant predictions to `checkpoints/<model>/per\_plant\_predictions.csv`.

5. Visualise suspicious plants

```
```bash
python visualize_suspicious.py
```
```

Renders keypoint overlays for plants with anomalous stem length estimates. Output images saved to `suspicious\_viz/`.

Results (EfficientNet-B3 - baseline)

| Mode | Overall MAE (m) | Overall RMSE (m) |
|---------------------------|-----------------|------------------|
| ----- | ----- | ----- |
| Single-view | 0.186 | 0.278 |
| Multi-view (75 views avg) | 0.183 | 0.274 |

> These are early baseline results. Negative R2 values indicate the models are not yet outperforming a mean predictor, largely due to high intra-class variance and single-image ambiguity. Multi-view aggregation provides a modest improvement.

Repository Structure

```

...

keypoints_pipeline/
+-- reg_model.py      # EfficientNetReg, ResNet34Reg, CustomCNNReg
+-- reg_dataset.py    # PlantRegDataset, augmentation, target building
+-- train_engine.py    # Shared training loop (optimizer, scheduler, checkpointing)
+-- train_efficientnet.py # EfficientNet-B3 entry point
+-- train_resnet34.py  # ResNet-34 entry point
+-- train_custom.py    # Custom CNN entry point
+-- train_all.py       # Run all three sequentially
+-- eval_reg.py        # Evaluation & metrics
+-- measure_plants.py  # 3D point cloud -> labels/keypoints
+-- project_keypoints.py # 3D keypoints -> 2D projections
+-- visualize_suspicious.py # Keypoint overlay renderer
+-- requirements.txt
+-- checkpoints/
    +-- efficientnet_b3/
    +-- resnet34/
    +-- custom_cnn/
...

---
```

Data Format Notes

****labels.json****

```

```json
{
 "0001": {
 "stem_length_m": 1.6445,
 "internode_lengths_m": [0.103, 0.115, ...],
 "leaf_lengths_m": [0.147, 0.949, ...]
 }
}
```

```

****keypoints_2d.json****

```

```json
{
 "plant_0001_rgb_001": {
 "keypoints": [[305, 210, 1], [306, 27, 0], ...],
 "elevation_deg": -30,
 "distance_m": 1.0
 }
}
```

```

Visibility flag: `1` = keypoint projects inside the image frame, `0` = occluded/out-of-frame.

License

See repository root for license information.

Both files are written:

requirements.txt - 7 packages: torch, torchvision, numpy, scipy, Pillow, plyfile, opencv-python (all with minimum version pins)

README.md - covers pipeline flow, dataset stats (520 plants / ~39k images / 75 views), all three models and their configs, install/usage instructions for every script, and the EfficientNet baseline results from eval_results.json

The eval results section honestly notes the negative R2 values (model not yet outperforming mean predictor) so reviewers have accurate expectations.

c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\README.md

save a pdf of this conversation

Bash Check if pandoc is available

IN

```
pandoc --version 2>/dev/null || echo "NOT_FOUND"
```

OUT

Exit code 1

Bash Check available Python PDF libraries

IN

```
python -c "import weasyprint; print('weasyprint OK')" 2>/dev/null || python -c "import fpdf; print('fpdf OK')"  
2>/dev/null || python -c "import reportlab; print('reportlab OK')" 2>/dev/null || echo "NO_PDF_LIB"
```

OUT

Exit code 1

Bash Install fpdf2 for PDF generation

IN

```
pip install fpdf2 --quiet 2>&1 | tail -1
```

OUT

Exit code 1

Bash Install fpdf2

IN

```
pip install fpdf2 2>&1 | tail -3
```

OUT

Exit code 1

Bash Install fpdf2 via pip3

IN

```
pip3 install fpdf2 2>&1 | tail -5
```

OUT

Exit code 1

Bash Check Python and pip availability

IN

```
python --version && pip --version
```

OUT

Exit code 127

Bash Locate Python executable on Windows

IN

where python 2>/dev/null; where python3 2>/dev/null; where py 2>/dev/null

OUT

Exit code 127

PowerShell

IN

```
{  
  "command": "Get-Command python, python3, py -ErrorAction SilentlyContinue | Select-Object Name, Source",  
  "description": "Find Python on Windows"  
}
```

OUT

(PowerShell completed with no output)

PowerShell

IN

```
{  
  "command": "Get-ChildItem \"C:\\Users\\sudanb\\anaconda3\\Scripts\\pip.exe\",  
  \"C:\\Users\\sudanb\\AppData\\Local\\Programs\\Python\\*\\python.exe\", \"C:\\ProgramData\\Miniconda3\\python.exe\"  
  -ErrorAction SilentlyContinue | Select-Object FullName",  
  "description": "Search common Python install locations"  
}
```

OUT

Exit code 1

PowerShell

IN

```
{  
  "command": "Get-ChildItem \"C:\\Users\\sudanb\" -Filter \"python.exe\" -Recurse -Depth 4 -ErrorAction  
  SilentlyContinue | Select-Object FullName -First 5",  
  "description": "Search for python.exe under user home"  
}
```

OUT

(PowerShell completed with no output)

PowerShell

IN

```
{  
  "command": "Get-ChildItem \"C:\\\" -Filter \"python.exe\" -Recurse -Depth 5 -ErrorAction SilentlyContinue |  
  Select-Object FullName -First 5",  
  "description": "Search for python.exe system-wide (depth 5)"  
}
```

OUT

Exit code 1

FullName

C:\gurobi952\win64\python37\bin\python.exe

C:\isaacsim\kit\python\python.exe

C:\Users\sudanb\Downloads\kit-sdk-publi...

PowerShell

IN

```
{  
  "command": "& \"C:\\isaacsim\\kit\\python\\python.exe\" -c \"import fpdf; print('fpdf OK')\" 2>&1\n&  
  \"C:\\isaacsim\\kit\\python\\python.exe\" -c \"import sys; print(sys.version)\" 2>&1",  
  "description": "Check isaacsim python and fpdf availability"
```



```

}
OUT
python.exe : Traceback (most recent call last):
At line:1 char:1
+ & "C:\isaacsim\kit\python\python.exe" -c "import fpdf; print('fpdf OK ..."
+ ~~~~~
+ CategoryInfo          : NotSpecified: (Traceback (most recent call last)
::String) [], RemoteException
+ FullyQualifiedErrorId : NativeCommandError

File "<string>", line 1, in <module>
ModuleNotFoundError: No module named 'fpdf'
3.11.13 (tags/v3.11.13-dirty:498b971, Jun  9 2025, 09:38:51) [MSC v.1912 64 bit (AMD64)]
PowerShell
IN
{
  "command": "& 'C:\isaacsim\kit\python\python.exe' -m pip install fpdf2 --quiet 2>&1 | Select-Object -Last
3",
  "description": "Install fpdf2 into isaacsim Python"
}
OUT
[notice] A new release of pip is available: 24.3.1 -> 26.1
[notice] To update, run: C:\isaacsim\kit\python\python.exe -m pip install
--upgrade pip
Good, fpdf2 is installed. Now I'll write a script to generate the PDF and run it.

Write c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\export_conversation.py
143 lines
"""Generate a PDF of the keypoints_pipeline conversation."""

from fpdf import FPDF
from fpdf.enums import XPos, YPos
import textwrap, os

CONV = [
  {
    "role": "User",
    "text": (
      "get context of the directory : keypoints_pipeline and generate a "
      "requirements.txt and readme.md for github push"
    ),
  },
  {
    "role": "Assistant",
    "text": (
      "Explored the keypoints_pipeline directory thoroughly.\n\n"
      "Pipeline overview:\n"
      " 1. measure_plants.py - extract stem/internode/leaf lengths from "
      "segmented PLY point clouds via Minimum Spanning Tree arc-length fitting.\n"
      " 2. project_keypoints.py - project 3D leaf-attachment keypoints onto "
      "2D RGB images using Intel RealSense D435 camera intrinsics (640x640, "
      "hfov=90deg, vfov=65deg).\n"

```

```

" 3. reg_dataset.py / reg_model.py / train_engine.py - build PyTorch "
"Dataset and train deep regression models (EfficientNet-B3, ResNet-34, "
"Custom CNN) to predict 32 plant-measurement targets from RGB images.\n"
" 4. eval_reg.py - evaluate single-view and 75-view-aggregated MAE, "
"RMSE, R2 metrics and export per-plant predictions to CSV.\n\n"
"Dataset: ~520 plants, 75 views each (~39 000 images), 640x640 PNG.\n"
"Targets: 1 stem length + 15 internode lengths + 16 leaf lengths = 32.\n"
"Loss: Masked MSE (only valid targets contribute).\n\n"
"Generated files:\n"
" * requirements.txt - torch>=2.0.0, torchvision>=0.15.0, numpy>=1.24.0,"
" scipy>=1.10.0, Pillow>=9.0.0, plyfile>=0.8.1, opencv-python>=4.7.0\n"
" * README.md - full GitHub-ready documentation covering pipeline flow,"
" dataset, models, installation, usage, baseline results, and directory "
"structure."
),
},
{
"role": "User",
"text": "save a pdf of this conversation",
},
{
"role": "Assistant",
"text": (
"Searched for PDF tooling (pandoc, weasyprint, fpdf, reportlab) - none "
"pre-installed. Located Python 3.11 at C:\\isaacsim\\kit\\python\\python.exe "
"and installed fpdf2. Generated this PDF via export_conversation.py."
),
},
},
]

```

```
OUT = os.path.join(os.path.dirname(__file__), "conversation.pdf")
```

```
MARGIN = 15
```

```
PAGE_W = 210
```

```
class PDF(FPDF):
```

```
    def header(self):
```

```
        self.set_font("Helvetica", "B", 9)
        self.set_text_color(120, 120, 120)
        self.cell(0, 8, "keypoints_pipeline - Session Conversation", align="C",
                  new_x=XPos.LMARGIN, new_y=YPos.NEXT)
        self.set_draw_color(200, 200, 200)
        self.line(MARGIN, self.get_y(), PAGE_W - MARGIN, self.get_y())
        self.ln(2)
```

```
    def footer(self):
```

```
        self.set_y(-13)
        self.set_font("Helvetica", "I", 8)
        self.set_text_color(150, 150, 150)
        self.cell(0, 8, f"Page {self.page_no()}", align="C")
```

```
pdf = PDF()
```

```
pdf.set_margins(MARGIN, MARGIN, MARGIN)
pdf.set_auto_page_break(auto=True, margin=15)
pdf.add_page()

# Title
pdf.set_font("Helvetica", "B", 16)
pdf.set_text_color(30, 30, 30)
pdf.cell(0, 10, "keypoints_pipeline - Conversation Log", align="C",
        new_x=XPos.LMARGIN, new_y=YPos.NEXT)
pdf.set_font("Helvetica", "", 9)
pdf.set_text_color(130, 130, 130)
pdf.cell(0, 6, "Generated 2026-05-04", align="C",
        new_x=XPos.LMARGIN, new_y=YPos.NEXT)
pdf.ln(6)

USERBG = (230, 240, 255)
ASSISTBG = (240, 250, 240)
LABELW = 22
BODY_W = PAGE_W - 2 * MARGIN - LABELW - 4

for turn in CONV:
    is_user = turn["role"] == "User"
    bg = USERBG if is_user else ASSISTBG
    label = turn["role"]
    body = turn["text"]

    # Measure text height
    pdf.set_font("Helvetica", "", 9)
    lines = []
    for para in body.split("\n"):
        if para.strip() == "":
            lines.append("")
        else:
            lines.extend(textwrap.wrap(para, width=90) or [""])
    line_h = 5
    box_h = len(lines) * line_h + 6

    # Background rect
    x0 = MARGIN
    y0 = pdf.get_y()
    if y0 + box_h > pdf.page_break_trigger:
        pdf.add_page()
        y0 = pdf.get_y()

    pdf.set_fill_color(*bg)
    pdf.rect(x0, y0, PAGE_W - 2 * MARGIN, box_h, style="F")

    # Role label
    pdf.set_xy(x0 + 2, y0 + 3)
    pdf.set_font("Helvetica", "B", 8)
    pdf.set_text_color(60, 60, 60)
    pdf.cell(LABELW, line_h, label)
```

```
# Body text
pdf.set_xy(x0 + LABELW + 2, y0 + 3)
pdf.set_font("Helvetica", "", 9)
pdf.set_text_color(30, 30, 30)
for i, line in enumerate(lines):
    if i > 0:
        pdf.set_x(x0 + LABELW + 2)
        pdf.cell(BODY_W, line_h, line, new_x=XPos.LMARGIN, new_y=YPos.NEXT)

pdf.ln(3)

pdf.output(OUT)
print(f"Saved: {OUT}")
PowerShell
IN
{
  "command": "& 'C:\\isaacsim\\kit\\python\\python.exe'"
  "C:\\Users\\sudanb\\Desktop\\CV_datasets\\keypoints_pipeline\\export_conversation.py\\",
  "description": "Run PDF export script"
}
```

OUT

Exit code 1

Traceback (most recent call last):

```
File "C:\\isaacsim\\kit\\python\\Lib\\site-packages\\fpdf\\fpdf.py", line 5681, in normalize_text
    return text.encode(self.core_fonts_encoding).decode("latin-1")
           ~~~~~~
```

UnicodeEncodeError: 'latin-1' codec can't encode character '\u2014' in position 19: ordinal not in range(256)

The above exception was the direct cause of the following exception:

Traceback (most recent call last):

```
File "C:\\Users\\sudanb\\Desktop\\CV_datasets\\keypoints_pipeline\\export_conversation.py", line 79, in <module>
    pdf.add_page()
```

```
File "C:\\isaacsim\\kit\\python\\Lib\\site-packages\\fpdf\\fpdf.py", line 1214, in add_page
    self.header()
```

```
File "C:\\Users\\sudanb\\Desktop\\CV_datasets\\keypoints_pipeline\\export_conversation.py", line 63, in header
    self.cell(0, 8, "keypoints_pipeline ? Session Conversation", align="C",
```

```
File "C:\\isaacsim\\kit\\python\\Lib\\site-packages\\fpdf\\fpdf.py", line 281, in wrapper
    return fn(*args, **kwargs)
           ~~~~~~
```

```
File "C:\\isaacsim\\kit\\python\\Lib\\site-packages\\fpdf\\deprecation.py", line 36, in wrapper
    return fn(*args, **kwargs)
           ~~~~~~
```

```
File "C:\\isaacsim\\kit\\python\\Lib\\site-packages\\fpdf\\fpdf.py", line 3910, in cell
    text = self.normalize_text(text)
           ~~~~~~
```

```
File "C:\\isaacsim\\kit\\python\\Lib\\site-packages\\fpdf\\fpdf.py", line 5683, in normalize_text
    raise FPDFUnicodeEncodingException(
```

fpdf.errors.FPDFUnicodeEncodingException: Character "?" at index 19 in text is outside the range of characters supported by the font used: "helveticaB". Please consider using a Unicode font.

Edit c:\\Users\\sudanb\\Desktop\\CV_datasets\\keypoints_pipeline\\export_conversation.py

Modified

Edit c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\export_conversation.py

Modified

Edit c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\export_conversation.py

Modified

PowerShell

IN

```
{  
  "command": "& 'C:\isaacsim\kit\python\python.exe'"  
  "C:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\export_conversation.py\"",  
  "description": "Run PDF export script"  
}
```

OUT

Exit code 1

Traceback (most recent call last):

```
File "C:\isaacsim\kit\python\Lib\site-packages\fpdf\fpdf.py", line 5681, in normalize_text  
    return text.encode(self.core_fonts_encoding).decode("latin-1")  
           ~~~~~~
```

UnicodeEncodeError: 'latin-1' codec can't encode character '\u2014' in position 24: ordinal not in range(256)

The above exception was the direct cause of the following exception:

Traceback (most recent call last):

```
File "C:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\export_conversation.py", line 137, in <module>  
    pdf.cell(BODY_W, line_h, line, new_x=XPos.LMARGIN, new_y=YPos.NEXT)  
File "C:\isaacsim\kit\python\Lib\site-packages\fpdf\fpdf.py", line 281, in wrapper  
    return fn(*args, **kwargs)  
           ~~~~~~
```

```
File "C:\isaacsim\kit\python\Lib\site-packages\fpdf\deprecation.py", line 36, in wrapper  
    return fn(*args, **kwargs)  
           ~~~~~~
```

```
File "C:\isaacsim\kit\python\Lib\site-packages\fpdf\fpdf.py", line 3910, in cell  
    text = self.normalize_text(text)  
           ~~~~~~
```

```
File "C:\isaacsim\kit\python\Lib\site-packages\fpdf\fpdf.py", line 5683, in normalize_text  
    raise FPDFUnicodeEncodingException(  
fpdf.errors.FPDFUnicodeEncodingException: Character "?" at index 24 in text is outside the range of characters
```

supported by the font used: "helvetica". Please consider using a Unicode font.

Read c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\export_conversation.py

Edit c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\export_conversation.py

Modified

Edit c:\Users\sudanb\Desktop\CV_datasets\keypoints_pipeline\export_conversation.py

Edit failed

Click to expand